ChatGPT

Deep Review of Graph-Conditioned Neural Decompilation of Dart AOT Binaries

Executive summary

The paper is **promising but not yet suitable for a Core A/A*** full research venue in its current form. My overall recommendation is **Reject** for a top-tier full paper, with a realistic path to resubmission after a substantial revision cycle. The main strengths are clear problem framing, a useful insistence on **functional correctness** rather than surface similarity, an interesting **graph-conditioned** encoder–decoder design for Dart AOT, and unusually candid discussion of several limitations, especially the gap between small specialised models and frontier zero-shot systems, the difference between single-model performance and pooled candidate coverage, and the incompleteness of the current artifact. filecite turn0file0

The main reasons this falls below a Core A/A* threshold are methodological and positioning-related rather than purely engineering-related. The evaluation remains heavily tied to a **small 154-task HumanEval-Dart benchmark**, the paper explicitly states that several results are **exploratory rather than seed-averaged**, some headline improvements come from **non-held-out pooled-arm and harvesting analyses**, and the current draft does not yet supply a submission-ready artifact because **weights, full environment manifests, and some command logs remain missing**. Those are not fatal for an early-stage study, but they are material weaknesses for ASE/ICSE/ICLR/EMNLP/CCS-level review. filecite turn0file0

There are also a handful of **clear or likely factual/citation problems**. Two stand out. First, the paper states that MultiPL-E includes **Dart** in its benchmark family, but the original MultiPL-E benchmark covers **18 languages** and does **not** include Dart. Second, the paper's discussion of SLaDe appears to conflate its evaluation setup with ExeBench, whereas the SLaDe paper itself describes evaluation on **AnghaBench** and SynthBench-style settings. Several bibliographic entries also appear inaccurate or incomplete, including the titles attached to MultiPL-E and Idioms. These are repairable, but they matter because the manuscript explicitly positions itself as a careful empirical and factual correction to prior decompilation evaluation practice.

My bottom-line judgement is therefore: **interesting work, worthwhile direction, potentially publishable after major strengthening, but not yet at the evidential, reproducibility, or reference-hygiene bar expected of a Core A/A* full paper**.

Factual accuracy and citation audit

The table below separates **clear factual errors**, **probable bibliographic errors**, and **publication-readiness issues that are not exactly factual errors but would matter in review**.

Location in submitted paper	Claim or issue	Assessment	Evidence	Why it matters	Suggested correction
p. 4, §2.2 Related Work	MultiPL-E is described as spanning “many programming languages, **including Dart** in its polyglot benchmark family.”	**Clear factual error**	MultiPL-E’s primary paper describes translation of HumanEval/MBPP into **18 languages**; the listed languages include Swift, Racket, Rust, Scala, etc., but **not Dart**. filecite turn0file0 [1]	This directly affects the novelty framing of HumanEval-Dart.	Rephrase to say MultiPL-E established the *general paradigm* of polyglot benchmark translation, while **HumanEval-Dart is a new Dart-specific adaptation**.
pp. 2–4, Related Work around SLaDe	The manuscript’s discussion implies SLaDe is part of an ExeBench-centred evaluation lineage.	**Likely factual mischaracterisation**	The SLaDe paper describes evaluation on **over 4,000 executable programs from AnghaBench** and contrasts itself with SynthBench-style settings. Its body states AnghaBench explicitly. filecite turn0file0 [2]	The distinction matters because AnghaBench, SynthBench, and ExeBench imply different realism and difficulty profiles.	State precisely which datasets each prior paper used; do not group SLaDe into an ExeBench bucket unless quoting a secondary source and explaining the mismatch.
Ref. [4], p. 17	“MultiPL-E: A scalable and **polyglot** approach to benchmarking neural code generation.”	**Probable bibliographic error**	The accessible primary record gives the title as **“MultiPL-E: A Scalable and Extensible Approach to Benchmarking Neural Code Generation.”** filecite turn0file0 [3]	Small on its own, but together with the Dart error it suggests the paper’s discussion of MultiPL-E was not checked against the primary source.	Replace with the canonical title and, ideally, DOI or stable publisher metadata.

Location in submitted paper	Claim or issue	Assessment	Evidence	Why it matters	Suggested correction
Ref. [7], p. 17	“Idioms: Neural decompilation with joint code and **type definition** prediction.”	**Probable bibliographic error**	The accessible primary paper is titled **“Idioms: Neural Decompilation With Joint Code and Type Prediction.”** filecite turn0file0 4	Reference hygiene matters because this paper is itself about careful evaluation and interpretation.	Use the exact published or preprint title.
Ref. [17], p. 18	“The CodeInverter suite ..., 2025.” No venue, DOI, arXiv ID, or repository.	**Unverifiable / incomplete citation**	I could not locate a primary record via targeted search, and the bibliography itself gives no stable identifier. filecite turn0file0	Readers cannot verify the claim that this is representative prior work.	Supply a verifiable identifier or remove the citation from critical novelty claims until one exists.
p. 14, §7 Artifact and Reproducibility Notes	The paper says anonymised repository metadata and self-identifying prior-work citations “will be scrubbed” upon acceptance.	**Submission-readiness problem**	The current draft already contains full author names on the title page and strongly self-identifying references [1] and [2]. filecite turn0file0	This is not a scientific error, but it makes the current draft unsuitable for double-anonymous review workflows common in several top venues.	Prepare a genuinely anonymised submission version and an anonymised artifact link at submission time, not after acceptance.

The factual mistakes are not catastrophic, but they are concentrated exactly in the section where the paper makes its strongest novelty claims. That weakens confidence in the paper’s “we know where this work sits in the literature” argument.

There is also a broader reference-hygiene issue. Several entries are preprints or informal resources without stable bibliographic metadata, which is sometimes unavoidable in a fast-moving area, but then the manuscript must be especially careful to quote titles, venues, and claims exactly. At present, that care is inconsistent.

Methodology, experimental design, and statistical assessment

The paper’s central experimental instinct is good: it treats **pass@k** as the main signal of semantic recovery, keeps CodeBLEU and compile metrics as diagnostics, and explicitly distinguishes single-model generalisation from pooled candidate coverage. That is well aligned with the original HumanEval framing, where pass@k is the standard estimator for functional correctness under repeated sampling, and it is also consistent with more recent concerns that token-similarity metrics are weak proxies for true program behaviour.

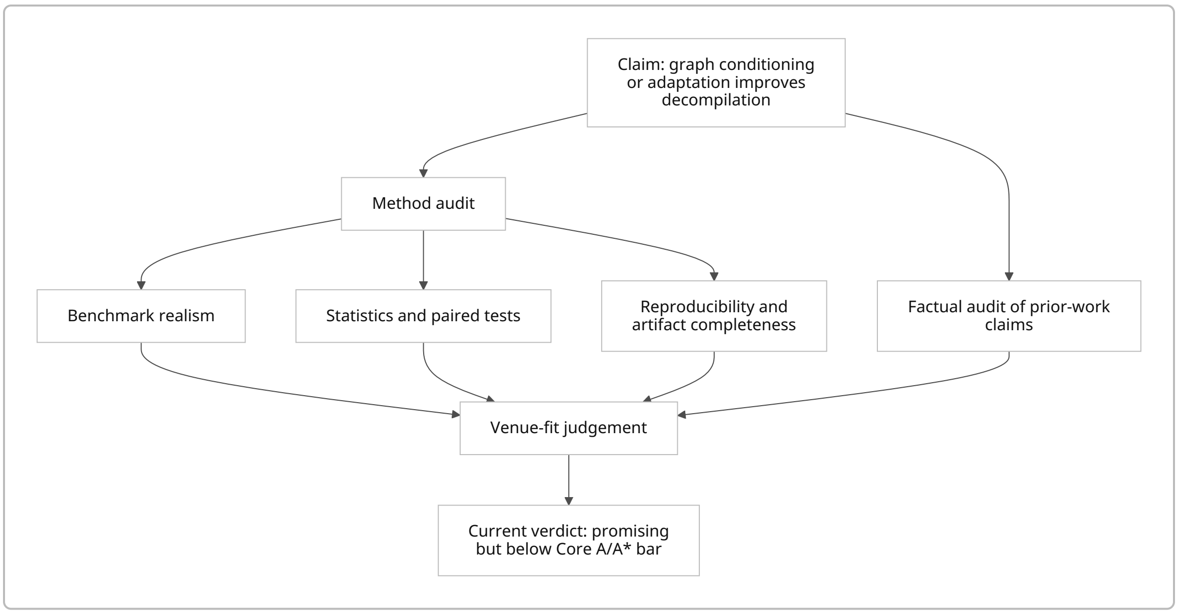
At the same time, the paper itself documents several weaknesses that become decisive at a Core A/A* bar. It admits that the reported runs are **“exploratory rather than seed-averaged”**, that confidence intervals are over **one archived generation pool** rather than retraining uncertainty, that **harvest and ensemble** results should be treated as optimisation/coverage analysis rather than clean generalisation, and that the benchmark size is **insufficient for deployment readiness**. Those are exactly the kind of limitations reviewers at top venues tend to turn into “interesting but not yet mature” decisions. fileciteturn0file0

A second issue is **benchmark external validity**. The entire argument is built primarily on **154 HumanEval-Dart tasks**, which are function-level and test-driven. That makes sense for a first benchmark, but it is still a long way from real Flutter/Dart AOT reverse engineering, where package structures, runtime conventions, object layouts, framework calls, cross-function context, and build-system artefacts matter. The paper implicitly acknowledges this by repeatedly retreating to “HumanEval-scale Dart AOT functions” and by listing larger Flutter/Dart corpora as future work. A top venue will likely ask why the stronger claims are made before the stronger benchmark exists.

A third issue is **contamination and task-identity leakage**. The paper itself proposes, as future work, an exact token-cap test and a **signature-only control** to quantify how much HumanEval task identity can be recovered from function signatures alone. That is not a minor clean-up item; it is central to whether the benchmark really measures decompilation rather than task retrieval or benchmark familiarity. Broader work on benchmark contamination has shown that popular public benchmarks, including HumanEval-class settings, are vulnerable even when overlaps are indirect or rephrased.

A fourth issue is that the paper mixes **two compile notions** in a way that will confuse readers. Table 1 uses a legacy 126-task compile/CodeBLEU metric; Table 2 switches to a 154-task **JIT/pass-harness aligned compile**; the text then explains why the latter is better aligned with pass@k and why the former can create misleading impressions. That logic is defensible, but the presentation is not reviewer-friendly. A paper should not require the reader to keep a latent “legacy compile vs aligned compile” ontology in mind to understand whether an arm improved or regressed. fileciteturn0file0

The statistical testing is mixed. Using task-paired **exact McNemar tests** for binary coverage changes is sensible, and the paper does a better job than many empirical ML papers in separating paired task gains/losses from headline pass@k values. But the inferential story is still incomplete because many pairwise comparisons are explored, the paper does not foreground a small preregistered set of primary hypotheses, and no repeated-seed uncertainty is reported. That makes the p-values useful as diagnostics, but not strong enough to support expansive comparative claims at top-tier level. fileciteturn0file0



Methodological fixes that would materially strengthen the paper

Problem	Why it is serious	Concrete fix
Benchmark may leak task identity via signature or public benchmark familiarity	Undermines causal claims about “decompilation ability”	Add **signature-only**, **docstring-free**, and **fresh post-cutoff real-world** evaluation slices; report them separately.
154 tasks and no real-world Flutter corpus	Limits external validity and venue significance	Add a second benchmark of real Dart/Flutter AOT functions with buildable package context, or at minimum a carefully time-split real-world holdout.
No seed-averaged training uncertainty	Makes cross-arm conclusions fragile	Repeat the main arms across multiple seeds and report variance for pass@1/pass@10 and paired gains/losses. filecite turn0file0
Mixed compile definitions	Confuses readers and can mislead interpretation	Put one compile definition in the main paper; move the alternate diagnostic compile view to an appendix. filecite turn0file0
Frontier comparison is not architecture-controlled	Hard to know whether the main story is scale, data, prompt design, or proprietary serving effects	Add stronger open-weight baselines, matched prompt/token budgets, and explicit “apples-to-apples” comparisons where possible.
Harvest/union results are not held-out generalisation	Easy for reviewers to see as benchmark gaming	Move such results into a clearly labelled **candidate-discovery** section and add held-out reranking/harvesting evaluation. filecite turn0file0

Novelty, significance, and fit for likely Core A/A* venues

The paper's novelty is **real but narrow**. The genuinely new elements are the **Dart AOT focus**, the integration of **graph-conditioned encoding** into this setting, the careful distinction between **metric mismatch** and **functional recovery**, and the explicit treatment of **multi-arm candidate coverage** as different from clean checkpoint generalisation. Those are useful contributions. However, the broader building blocks are already active and increasingly strong in the literature: GraphCodeBERT already established data-flow-aware code representations, DIRE and DIRTY already showed learned recovery of decompiler-lost semantic artefacts, LLM4Decompile already built an open decompilation LLM family with executability emphasis, and Nova already demonstrated structure-aware assembly modelling with strong gains on decompilation-related tasks.

The significance is therefore not "a new decompilation paradigm", but rather "a careful empirical study showing how far a graph-conditioned small-model Dart decompiler currently gets, and where it still fails against frontier zero-shot models". That is publishable in principle, but for a top-tier venue the study needs either **more decisive evidence** or **stronger novelty**. Right now it has neither in sufficient degree.

Comparison with representative top-tier papers

Paper	Venue	Core-A/ A* relevance	Main contribution	Method style	Data / evaluation	Headline result	W r s p
DIRE	ASE 2019	Top SE venue	Variable-name recovery from decompiled code	Neural model over lexical + structural decompiler information	164,632 x86-64 binaries; exact-name recovery	Up to **74.3%** exact variable-name recovery	S r s e s r s t y r c e i s

Paper	Venue	Core-A/ A* relevance	Main contribution	Method style	Data / evaluation	Headline result	W r s p
GraphCodeBERT	ICLR 2021	Top ML venue	Data-flow-aware pre-training for code representation	Graph-guided Transformer pre-training	CodeSearchNet; four downstream tasks	SOTA across four code tasks	Y c i c a t l s s t e r
DIRTY	USENIX Security 2022	Top security venue	Joint recovery of variable names and types from decompiler output	Learned semantic augmentation of decompiler output	GitHub-mined C corpus	**66.4%** name recovery, **75.8%** type recovery	D h a t f c r e p c e f r
LLM4Decompile	EMNLP 2024	Top NLP venue	Open LLM family for decompilation with executability-oriented evaluation	Large decompilation-specific LLM training	HumanEval + ExeBench / Decompile-Eval	Over **100%** relative re-executability gains vs GPT-4o/ Ghidra in abstract; 21% execution success reported in intro	T c p F y c r l c e r r s h e a

Paper	Venue	Core-A/ A* relevance	Main contribution	Method style	Data / evaluation	Headline result	W r s p
Nova	ICLR 2025	Top ML venue	Hierarchical attention and contrastive learning for assembly	Assembly-native generative LM	Decompilation + binary similarity tasks	**14.84–21.58 p.p.** decompilation gains; **6.17%** Recall@1 over prior work	T h r f “ p r t r r

Relative to these papers, the current submission looks strongest as an **empirical evaluation and positioning paper** rather than as a pure architecture paper. That suggests its best top-tier path is via a venue that rewards careful empirical SE evidence and artifact quality. Among likely homes, **ASE** appears more plausible than **ICSE**, while **EMNLP** or **ICLR** would demand a stronger modelling or benchmark contribution, and **CCS/S&P** would require a much more substantive security case with real-world binary-analysis value. [10]

Venue-fit assessment

Likely venue family	Fit to topic	Current readiness	Main blockers
ASE / software engineering	Good topical fit: automated code analysis, code intelligence, empirical evaluation	**Borderline to weak**	Missing submission-ready artifact, small benchmark, non-anonymised draft, limited external validity. [11]
ICSE / premier SE	Conceptually relevant, but bar is very high on significance and evidence	**Weak**	Needs broader real-world significance and more mature evidence package. [12]
EMNLP / NLP-for-code	Reasonable because decompilation is framed as code generation	**Moderate fit, weak readiness**	Current modelling novelty seems too incremental versus LLM4Decompile/Nova; benchmark too small. [13]
ICLR / ML	Only if sold as a genuinely new structure-aware assembly modelling method	**Weak**	Novelty delta versus GraphCodeBERT/Nova is not yet large enough. [14]
CCS / S&P / security	Reverse-engineering motivation is present	**Weak**	No strong real-world security dataset, threat model, or analyst-facing analysis yet; dual-use discussion is too light. [15]

Reproducibility, ethics, and writing quality

The artifact section is commendably frank, and that honesty helps. The paper claims the working repository already contains code, data manifests, prediction files, sweep summaries, candidate-level CSVs, frontier prompts, model identifiers, prompt hashes for some ablations, and a pinned **local verification environment** with regression checks. That is good practice. However, the same section explicitly confirms that **checkpoint weights are not archived**, **SHA-256 hashes are missing**, the **pod-side training environment is not fully pinned**, and some reranking arms were run interactively and are still being reconstructed from logs. That is a substantial reproducibility gap by top-tier standards. filecite turn0file0

A strong replication package is not optional here because the paper's main scientific value lies in nuanced empirical distinctions: legacy compile vs aligned compile, single-arm vs pooled-arm coverage, v1 vs v2 frontier ablations, and partial gains that are often non-significant. If reviewers cannot inspect the exact runs, prompts, and candidate pools, the paper becomes much less persuasive. That concern is reinforced by broader evidence from software-engineering artifact studies showing that artifact sharing is now standard practice in premier SE venues, while true executability and reproducibility remain difficult in practice.

The ethics section is directionally correct but too brief for the paper's domain. The paper notes that neural decompilation supports legitimate reverse engineering and defensive security work while also lowering the cost of analysing proprietary binaries, and it says releases should avoid proprietary binaries and be framed for authorised analysis. That is a reasonable baseline, but for a paper explicitly about decompilation of proprietary or security-relevant binaries, top-tier venues may expect a more concrete abuse analysis, clearer release boundaries, and stronger discussion of how benchmark-only release differs from capability release. filecite turn0file0

The writing quality is generally solid, but the paper is **overloaded with caveat-management**. It often says the right cautious thing, yet the accumulation of labels, controls, caveats, terminology distinctions, and archived run names makes the paper harder to read than it should be. The manuscript would benefit from moving more run-traceability detail into an appendix and giving the main paper a cleaner narrative arc. filecite turn0file0

Reproducibility checklist

Item	Status in current draft	What is still required for a top-tier submission
Source code	**Partial-to-strong** in working repo	Public anonymised submission link; precise commit hash; licence. filecite turn0file0
Data manifests	**Present**	Public manifests plus exact train/val/test provenance and cutoffs. filecite turn0file0
Raw predictions / candidate CSVs	**Present**	Public release and schema documentation. filecite turn0file0
Exact prompts / model IDs for frontier runs	**Present**	Public archive with timestamps and provider metadata. filecite turn0file0
Checkpoint weights / LoRA adapters	**Missing**	Release all adapters or provide a justified access mechanism. filecite turn0file0

Item	Status in current draft	What is still required for a top-tier submission
Hashes of checkpoints and datasets	**Missing**	SHA-256 or equivalent for every released artifact. filecituturn0file0
Environment manifest	**Partial**	Pin both local and pod-side training/inference environments. filecituturn0file0
Full run-command log	**Partial**	Finish reconstruction for all interactive reranking / repair arms. filecituturn0file0
Multiple seeds	**Missing**	Repeat the main reported arms across seeds. filecituturn0file0
Anonymised artifact	**Missing at submission**	Provide an anonymised repository during review, not only after acceptance. filecituturn0file0

Suggested edits for clarity and tone

Current pattern	Why it hurts clarity	Better editorial strategy
Too many archived run labels in the main flow	Makes the paper read like an experiment log	Keep descriptive names in the main text; move literal archive labels to appendix tables only. filecituturn0file0
Multiple definitions of compile spread across sections	Readers cannot easily track what changed	Introduce a one-paragraph "evaluation conventions" box early, and use one main compile metric in the paper body. filecituturn0file0
Caveats sometimes appear after the claim	Reviewers may perceive retreat from overstatement	Front-load the scope for every major claim in one sentence. filecituturn0file0
Frontier comparison narrative is longer than specialised-paper contribution narrative	Risks making the paper sound like "small models lose to bigger models"	Reframe around the actual contribution: **what specialised deployable systems can and cannot yet recover**, and why. filecituturn0file0
Related work contains a few factual slips	Erodes trust	Re-audit every cited paper against the primary record before resubmission.

Prioritised revisions for the authors

The most important revisions are the following.

First, **repair the literature audit**. Correct the MultiPL-E/Dart statement, clean up the SLaDe dataset description, fix the bibliographic titles and identifiers, and verify every reference against the primary record. This is the fastest high-leverage improvement because it immediately raises reviewer confidence.

Second, **strengthen the benchmark story**. Add the signature-only contamination control the paper already proposes, and, if at all possible, add a second evaluation slice with more realistic Flutter/Dart

AOT functions. Without that, the paper will continue to look benchmark-bound and only partially externally valid.

Third, **upgrade the empirical rigour**. Repeat the principal arms across training seeds, separate primary hypotheses from exploratory analyses, and move all pooled-harvest/union results into a clearly labelled candidate-discovery section. The current honesty is a strength; the next step is turning that honesty into a design that is unambiguously review-proof. filecite turn0file0

Fourth, **make the artifact submission-ready**. Release anonymised code, prompts, data manifests, run logs, adapter weights or at least all reproducible training outputs, full environment manifests, and hashes. For a paper whose principal contribution is empirical evaluation and interpretation, this is not peripheral.

Fifth, **tighten the venue-specific pitch**. If targeting a software-engineering venue, foreground empirical lessons about metric choice, decompilation evaluation, and deployable candidate selection. If targeting an NLP/ML venue, strengthen the model novelty. If targeting security, add analyst-oriented real-world case studies and a stronger responsible-release section. At present the paper tries to speak to all three audiences, and that dilutes its impact with each. [10]

On balance, I would encourage the work to continue, because the topic is worthwhile and the authors are already asking many of the right questions. But for a Core A/A* full paper, the manuscript currently reads as **an insightful intermediate report rather than a finished top-tier research article**. filecite turn0file0

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[1] [3] https://arxiv.org/abs/2208.08227
https://arxiv.org/abs/2208.08227

[2] https://arxiv.org/abs/2305.12520
https://arxiv.org/abs/2305.12520

[4] https://arxiv.org/abs/2502.04536
https://arxiv.org/abs/2502.04536

[5] https://arxiv.org/abs/1909.09029
https://arxiv.org/abs/1909.09029

[6] https://arxiv.org/abs/2009.08366
https://arxiv.org/abs/2009.08366

[7] https://arxiv.org/abs/2108.06363
https://arxiv.org/abs/2108.06363

[8] https://arxiv.org/abs/2403.05286
https://arxiv.org/abs/2403.05286

[9] https://arxiv.org/abs/2311.13721
https://arxiv.org/abs/2311.13721

[10] [11] https://en.wikipedia.org/wiki/International_Conference_on_Automated_Software_Engineering
https://en.wikipedia.org/wiki/International_Conference_on_Automated_Software_Engineering

[12] https://en.wikipedia.org/wiki/International_Conference_on_Software_Engineering
https://en.wikipedia.org/wiki/International_Conference_on_Software_Engineering

[13] https://en.wikipedia.org/wiki/Empirical_Methods_in_Natural_Language_Processing
https://en.wikipedia.org/wiki/Empirical_Methods_in_Natural_Language_Processing

[14] https://en.wikipedia.org/wiki/International_Conference_on_Learning_Representations
https://en.wikipedia.org/wiki/International_Conference_on_Learning_Representations

[15] https://en.wikipedia.org/wiki/IEEE_Symposium_on_Security_and_Privacy
https://en.wikipedia.org/wiki/IEEE_Symposium_on_Security_and_Privacy